

Daniel Thédié

Research Software Engineer

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Education and Employment

Research Software Engineer

2025–present

University of Edinburgh, Scotland

- Maintenance and development of the **BioDare2** data repository and analysis platform
 - Front- and back-end development
- Development of a data management system for the **Ambient-BD project**
 - Data management and analysis
 - User interface design

Post-doctoral research associate

2020–2024

University of Edinburgh, Scotland

- *In-vivo* dynamics of the RecBCD repair complex upon exposure of *E. coli* to antibiotics
 - Tracking of single fluorescently labelled proteins
 - Automated image acquisition, management and analysis of large datasets
- Development of a Python package (**PyBerries**) for data analysis in the lab
 - Produce stable, reliable code
 - Distribute package and provide updates to multiple users
- Maintenance and supervision
 - Regular microscope maintenance
 - Training for new users
 - Supervision of Master students

Ph.D. Student, Super-resolution microscopy

2016–2019

Institute of Structural Biology (IBS), Grenoble, France

Characterisation of photoconvertible fluorescent proteins for advanced single-molecule localisation microscopy

- In-vitro photophysical characterisation of fluorescent proteins
- Quantitative PALM (PhotoActivated Localisation Microscopy)
- Data analysis and simulation (Matlab)
- Regular maintenance and upgrading of a home-built PALM/STORM microscope

Biotechnology engineering student

2013–2016

Ecole Supérieure de Biotechnologie de Strasbourg (ESBS), France

Awarded an Engineering degree in Biotechnology and a Master's degree in structural biology

- Advanced training in varied disciplines including genetic engineering, programming, chemistry, mathematics, statistics

Internships

- **European Molecular Biology Laboratory (EMBL), Grenoble, France (6 months)**
Characterised the epigenetic regulator of gene expression p300 bound to several of its ligands
- **Institute of Genetics, Molecular and Cellular Biology (IGBMC), Strasbourg, France (2 months)**
Designed procedures for an automation platform to perform high-throughput pull-down and ATPase activity assays

- **Institute of Genetics, Molecular and Cellular Biology (IGBMC), Strasbourg, France (2 months)**
Applied bioinformatics tools to the structural study of a eukaryotic protein complex
- **Centogene AG, Rostock, Germany (3.5 months)**
Improved the genetic testing procedure for two tumor suppressor genes

Bachelor student in biology

2010–2013

Cergy-Pontoise University, France

Ranked first out of 131 students for the degree

- Core training in cellular and molecular biology
- Erasmus year at Copenhagen University, Denmark

Internships

- **Copenhagen University, Denmark (3 months)**
Studied the influence of a toxin on mammalian cell viability markers
- **Pierre and Marie Curie University, Paris, France (1 month)**
Studied the influence of glucose concentration on the *in-vitro* production of red blood cells

Software

Active Development

BioDare2 (Angular, TypeScript, Java)

Maintainer since 2025. Data repository and analysis platform for circadian data (biodare2.ed.ac.uk)

Ambient downloader (Python)

Download and manage sleep data from radar devices

REDCap downloader (Python)

Download and manage patient data from the REDCap platform (Ambient-BD project)

nocturn (R, R shiny)

Process and visualise sleep data. Includes a programmatic interface and an **R Shiny app**

svgedit (R)

Create reproducible, complex figure layouts by inserting elements into svg files (available on **CRAN**)

Past projects

PyBerries (Python)

Analyse quantitative data from microscopy images

Teaching

Workshop on git and GitHub (0.5 day)

2022, 2026

Internal

Basics of using git for code management, good coding practices ([workshop materials](#)).

Data anonymisation workshop (0.5 day)

2025

Edinburgh Circadian Mental Health Conference

Introduction to data anonymisation for researchers from different disciplines and career stages.

Workshop on the BACMMAN Fiji plugin (2 days)

2024

Internal lab training

Performing image analysis tasks (segmentation, image quantification) based on the participants' own data, troubleshooting

Teaching assistant, cell biology practicals (2 weeks)

2018, 2019

Grenoble-Alpes University

Helped students conceive experimental projects around mammalian cell culture and fluorescence microscopy. Marked oral presentations.

Supervision of Masters students

2023 - Effects of a DNA-damaging antibiotic on *E. coli* replication forks

2022 - Using deep-learning for bacterial cell segmentation

2018 - Structural dynamics of a phototransformable fluorescent protein at cryogenic temperature

2018 - Benchmarking methods for complex stoichiometry determination by PALM microscopy

Publications

First author

BioRxiv, 2025 (DOI)

Spatio-temporal coordination of the DNA Double-Strand Break repair machinery in *Escherichia coli*

PhD thesis, 2019 (DOI)

Characterisation of photoconvertible fluorescent proteins for single-molecule localisation microscopy

Journal of Physical Chemistry Letters, 2017 (DOI)

Photoswitching of green mEos2 by intense 561 nm light perturbs efficient green-to-red photoconversion in localization microscopy

Co-authorship

Nucleic Acids Research, 2025 (DOI)

In vivo single-molecule imaging of RecB reveals efficient repair of DNA damage in *Escherichia coli*

Wellcome Open Research, 2025 (DOI)

Ambient and passive collection of sleep and circadian data in bipolar disorder to understand symptom trajectories and clinical outcomes (AMBIENT-BD): Study protocol

Journal of Cell Science, 2024 (DOI)

The crucial role of bioimage analysts in scientific research and publication

Journal of the American Chemical Society, 2023 (DOI)

Photophysical studies at cryogenic temperature reveal a novel photoswitching mechanism of rsEGFP2

Journal of Physical Chemistry Letters, 2022 (DOI)

mEos4b Photoconversion Efficiency Depends on Laser Illumination Conditions Used in PALM

Journal of the American Chemical Society, 2020 (DOI)

Mechanistic investigations of green mEos4b reveal a dynamic long-lived dark state

Nature Methods, 2019 (DOI)

Mechanistic investigation of mEos4b reveals a strategy to reduce track interruptions in sptPALM

Conferences and Workshops

Circadian Mental Health innovation summit (2026, Edinburgh, UK)

Invited oral presentation

Diversity in the circadian, neuroendocrine and neurobehavioural effects of light in humans (2025, On-line)

Invited oral presentation

RSEcon (2025, Coventry, UK)

Participant

MetaScience (2025, London, UK)

Oral presentation

Images to Knowledge (2024, Milan, Italy)

Poster

Single-Molecule Bacteriology III (2024, De Vere Cotswolds Water Park, UK)

Oral presentation

Efficiently Communicating BioImage Analysis Workshop (2024, Buxted, UK)

Competitively selected to participate, oral presentation

Crick BioImage Analysis Symposium (2023, London, UK)

Poster

Single-Molecule Bacteriology II (2022, Didcot, UK)

Poster

EMBO symposium - New Approaches and Concepts in Microbiology (2021, Online)

Poster

Picoquant workshop (2019, Berlin, Germany)

Poster

Quantitative BioImaging (2019, Rennes, France)

Poster

Winter school on fluorescent markers for microscopy (2018, Les Houches, France)

Participant

Quantitative BioImaging (2018, Göttingen, Germany)

Oral presentation

Single Molecule Localisation Microscopy Symposium (2017, London, UK)

Poster

Outreach activities

Edinburgh Science Festival

2023

Engaging with families on the topics of microscopy, bacteria and antimicrobial resistance

Member of a young scientists committee

2018-2019

Grenoble, France

Organising research talks from external speakers

- Find speakers and invite them to give a talk about their research
- Help organise their travel and accommodation
- Welcome them to the institute, introduce their talk and schedule one-to-one discussions with group leaders

PhD work presentation in a high-school

2018-2019

Grenoble, France

Presenting fluorescence microscopy to 17-18 year-olds and answering their questions about scientific research